

A candidate counterexample to Question 4.1 in A note on deterministic zombies

github.com/graph-theory-AI — machine-generated candidate writeup

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Abstract

A 59-vertex cactus requires at least three zombies, but adding one leaf produces a graph requiring exactly two. This candidate disproof was generated by GPT-6 Astra in a single model pass for the Graph Theory AI project. It was selected from the campaign because the model marked it `would_publish: true`; it has not been checked by the adversarial referee, a human mathematician, or a novelty search.

1 The problem and the claimed result

This document concerns [Question 4.1](#) from *A note on deterministic zombies*.

1.1 Problem statement

Question. For every graph G , and for a graph G' obtained from G by successively adding vertices of degree 1, does it always hold that $z(G') = z(G)$?

1.2 Catalog context

This question asks whether attaching pendant vertices (leaves) to a graph leaves its zombie number unchanged. It can be interpreted as asking whether zombies can benefit from waiting at a designated position before activating at the start of the game.

1.3 Claimed result

Model verdict: disproved (confidence: high).

A 59-vertex cactus requires at least three zombies, but adding one leaf produces a graph requiring exactly two.

The model's own caveat: Uses the standard compulsory-move rules, with zombies moving before the survivor; no computational or literature-status claim is made.

2 Model writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

2.1 A counterexample

Zombies choose their initial positions first, followed by the survivor. On each turn, every zombie must move along an edge decreasing its distance to the survivor; the survivor then moves to a neighbor or stays. Capture occurs upon coincidence.

The counterexample is a cactus consisting of three cycles attached to a path. The proof below is entirely analytic.

2.1.1 1. Construction

Start with the path

$$v_0 v_1 \cdots v_{10}.$$

Attach, using otherwise disjoint vertices,

- a cycle A of length 25 at v_0 ;
- a cycle B of length 17 at v_2 ;
- a cycle C of length 9 at v_4 .

There are no other edges. Call the resulting graph G . Thus

$$|V(G)| = 11 + 24 + 16 + 8 = 59.$$

Let H be obtained by adding a single new vertex v_{11} , adjacent only to v_{10} . I will prove

$$z(G) \geq 3 \quad \text{and} \quad z(H) = 2.$$

Here and below, each cycle includes its attachment vertex.

2.1.2 2. A certificate for perpetual rotation around an odd cycle

Let Q contain a cycle of length

$$n = 2h + 1, \quad h \geq 2,$$

attached to the rest of Q only at its vertex 0. Choose an orientation and label its vertices by $\mathbb{Z}_n$ in that orientation.

For a zombie's initial position x , define its **phase**

$$p(x) = \begin{cases} x, & x \text{ lies on the cycle, using its cyclic coordinate;} \\ -d_Q(x, 0) \pmod{n}, & x \text{ lies outside the cycle.} \end{cases}$$

2.1.2.1 Rotation lemma If there is a coordinate q such that

$$q - p(x_i) \pmod{n} \in \{2, 3, \dots, h\} \quad (1)$$

for every zombie i , then the survivor can evade forever by starting at q and moving one step in the chosen orientation every turn.

In particular, two zombies can be evaded this way whenever

$$\|p(x_1) - p(x_2)\|_n \leq h - 2, \quad (2)$$

where $\|a\|_n$ denotes cyclic distance to 0 modulo n .

2.1.2.2 Proof First consider a zombie initially on the cycle. Condition (1) puts it a forward distance between 2 and h behind the survivor. Its unique shortest route is in the survivor's direction. After the zombie and survivor each move, this forward distance is unchanged.

Now consider a zombie outside the cycle, initially at distance D from the attachment vertex. While the survivor stays on the cycle, every shortest route from that zombie to the survivor passes through 0. Consequently, the zombie decreases its distance to 0 on every move and reaches 0 on its D -th move.

Immediately before that landing move, the survivor is at

$$q + D - 1 \pmod{n}.$$

Writing $e = q + D \pmod{n}$, condition (1) gives $2 \leq e \leq h$. Thus the survivor is not at 0 when the zombie lands. After the survivor's response, the forward gap is e , again between 2 and h , and remains constant thereafter.

For the final assertion, two residues can satisfy (1) simultaneously exactly when their difference has a representative in

$$[-(h-2), h-2],$$

because this is the set of differences of two integers in $\{2, \dots, h\}$. $\square$

For reference, the sufficient bounds in (2) are

Cycle length	h	$h-2$
25	12	10
17	8	6
9	4	2

2.1.3 3. Two zombies do not suffice on G

Consider any two initial zombie positions x, y .

Every vertex outside A has distance at most 10 from v_0 . Indeed:

- the path ends at v_{10} ;
- a vertex of B is at distance at most $2 + 8 = 10$ from v_0 ;
- a vertex of C is at distance at most $4 + 4 = 8$ from v_0 .

We distinguish all three possibilities for membership in A .

2.1.3.1 Case 1: Both zombies start outside A Put

$$D_x = d_G(x, v_0), \quad D_y = d_G(y, v_0).$$

Both numbers lie in $\{1, \dots, 10\}$, so

$$|D_x - D_y| \leq 9.$$

Their phases on A are $-D_x, -D_y$ modulo 25. Their cyclic separation is therefore at most 9, which is below the rotation lemma's bound 10. The survivor can rotate around A forever.

2.1.3.2 Case 2: Both zombies start in A Let

$$a = d_G(x, v_0), \quad b = d_G(y, v_0), \quad \delta = |a - b|.$$

Since A has length 25,

$$0 \leq \delta \leq 12.$$

The zombies' distances to v_2 are $a + 2, b + 2$, and their distances to v_4 are $a + 4, b + 4$. Thus their phase separation is represented by δ on both B and C .

If

$$\delta \in \{0, 1, \dots, 6, 11, 12\},$$

then

$$\|\delta\|_{17} \leq 6,$$

so the survivor can rotate around B .

Otherwise $\delta \in \{7, 8, 9, 10\}$, and then

$$\|\delta\|_9 \leq 2.$$

The survivor can instead rotate around C .

2.1.3.3 Case 3: Exactly one zombie starts in A Suppose $x \in A$ and $y \notin A$. Put

$$t = d_G(x, v_0) \in \{0, \dots, 12\}, \quad D = d_G(y, v_0) \in \{1, \dots, 10\}.$$

Orient A so that x has coordinate $-t \pmod{25}$. The two phases on A are then $-t, -D$, whose difference is represented by

$$D - t \in [-11, 10].$$

Except when

$$(D, t) = (1, 12),$$

we have $|D - t| \leq 10$, and the rotation lemma gives perpetual evasion on A .

In the exceptional case, $y = v_1$, since this is the only vertex outside A at distance 1 from v_0 . The distances of the zombies to the attachment vertex v_2 of B are

$$d_G(x, v_2) = 12 + 2 = 14, \quad d_G(y, v_2) = 1.$$

Their phase difference on B is 13 modulo 17, with cyclic distance

$$\|13\|_{17} = 4 \leq 6.$$

The survivor can rotate around B .

These cases cover every pair of initial positions. Therefore

$$\boxed{z(G) \geq 3.}$$

2.1.4 4. A delayed pair that captures on a pendant cycle

The following observation supplies the winning strategy on H .

2.1.4.1 Capture lemma Suppose a cycle of length $2d + 3$, with $d \geq 1$, is attached to the rest of a graph only at its root r . Suppose, immediately before a zombie turn,

- one zombie is at r ;
- another zombie is outside the cycle at distance d from r ;
- the survivor is on the cycle.

Then these two zombies can force capture.

2.1.4.2 Proof Label the root 0. Choose the orientation in which the survivor's initial coordinate p satisfies

$$1 \leq p \leq d + 1.$$

If $p = 1$, the root zombie captures immediately. Assume otherwise.

The root zombie follows the survivor in this orientation. Consider a play avoiding capture, including capture on the following zombie turn. After t complete rounds, write the first zombie's coordinate as t and the survivor's unwrapped coordinate as u_t .

The forward gap

$$g_t = u_t - t$$

cannot increase: on a survivor move it changes by 0, -1 , or -2 , accounting for the preceding zombie move. A gap at most 1 results in immediate or next-turn capture. Hence, while evasion continues,

$$2 \leq g_t \leq p \leq d + 1. \quad (3)$$

These bounds also ensure that the first zombie's move in the chosen direction is always its unique shortest-path move.

For $0 \leq t \leq d$, (3) gives

$$0 < u_t \leq t + d + 1 \leq 2d + 1 < 2d + 3.$$

Thus the survivor cannot reach the attachment vertex during these rounds and cannot leave the cycle.

The outside zombie consequently reaches the root after exactly d moves. After the survivor's response on that round,

$$d + 2 \leq u_d \leq 2d + 1.$$

The zombies are now at coordinates d and 0. They bracket the survivor on the arc from d forward to 0, and the two distances along that arc are

$$u_d - d \leq d + 1, \quad 2d + 3 - u_d \leq d + 1.$$

Both are shortest-path distances.

The zombies now move inward along this arc. Each of these two distances is nonincreasing after a complete round, so the inward moves remain legal. The occupied endpoints prevent escape through the root, and the intervening arc loses two edges on every zombie turn. Capture follows. $\square$

2.1.5 5. Two zombies win on H

Place the zombies at

$$v_0 \quad \text{and} \quad v_{11}.$$

Project every vertex of an attached cycle to its attachment vertex on the path

$$P = v_0 v_1 \cdots v_{11}.$$

Until a zombie reaches the attachment vertex of the cycle currently containing the survivor, both zombies move along P toward the survivor's projection. These are legal shortest-path moves.

The survivor cannot cross an occupied path vertex. Consequently, unless capture or such a cycle-root contact occurs, the two zombies' positions after t moves are

$$v_t, \quad v_{11-t},$$

and the survivor's projection remains strictly between them. This shrinking interval cannot support evasion indefinitely.

It remains to handle the first cycle-root contact. All attachment indices are 0, 2, 4, so this first contact is made by the zombie coming from v_0 . If the relevant root is v_r , the contact occurs after r zombie moves, with positions

$$v_r, \quad v_{11-r}.$$

The distance of the second zombie from the root is

$$d = 11 - 2r.$$

The three possibilities are

r	d	Length of the attached cycle
0	11	$25 = 2 \cdot 11 + 3$
2	7	$17 = 2 \cdot 7 + 3$
4	3	$9 = 2 \cdot 3 + 3$

If the survivor is actually at the root, it is captured. Otherwise it lies inside that cycle and cannot leave through the occupied root on its ensuing response. The capture lemma therefore applies in every case.

Thus

$$z(H) \leq 2.$$

Conversely, one zombie can always be evaded on A : in the rotation lemma, a single phase can always be accommodated by choosing $q = p + 2 \pmod{25}$. This applies whether the zombie starts inside or outside A . Hence

$$\boxed{z(H) = 2.}$$

2.1.6 6. Conclusion

The explicitly constructed graphs satisfy

$$\boxed{z(H) = 2 < 3 \leq z(G),}$$

although H is obtained from G by adding just the leaf v_{11} adjacent to v_{10} .

Thus **zombie number is not invariant under leaf attachment**, even for cactus graphs.

The mechanism is precisely an initial-delay advantage: the new leaf delays the rear zombie by one move, making its arrival gap at each of the three cycle roots large enough to approach the survivor from the opposite direction. The lower-bound argument shows that no alternative pair of starting vertices in G can achieve a global win.

Every pair of starting positions in G and every survivor start in H is covered above. Determining the exact value of $z(G)$ beyond $z(G) \geq 3$ is unnecessary for the counterexample.

A Provenance and review status

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim as unverified.

Catalog id	2008.03587__00
Catalog entry	Question 4.1
Source corpus	arxiv
Campaign leg	selected second attempts (attacks_retry)
Source paper / entry	A note on deterministic zombies
Model verdict	disproved (confidence: high)
Selection criterion	would_publish: true and verdict proved or disproved
Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_retry/2008.03587__00/output.md
Independent review	None
arXiv	arXiv:2008.03587
Previous attempt	attacks/2008.03587__00: gpt-5.6-sol verdict partial
Model's caveats	Uses the standard compulsory-move rules, with zombies moving before the survivor; no computational or literature-status claim is made.

The generated Markdown and LaTeX sources for this PDF are stored under `to_review_astra/src/2008.03587__00/`.